👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 6 Ignition (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 4: Ignition Handout, iCADEMY Bronze 7 Magneto Ignition Operation”)	Level 2 piece
Ignition Handout (title only)	Deck + notes + quiz (team-written)
iCADEMY Bronze 7 Magneto Ignition	Our own slides 'The Magneto' (no STIHL content)
Class slides	Deck: 15 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 9 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Show a real flywheel and armature side by side
- Spark plug passed around (clean, cold)

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Ignition Handout: use the notes as the handout; same points
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Ignition Handout**: Title only in Canvas; our notes and quiz are team-written.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Spark Plug & Spark Check (Level 2)

Small Engines Overhaul · Unit 6 · goes with the class “Ignition Handout” (lab points). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Ignition Handout”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Task card instead of written directions
- Sentence starters on the reflection
- Partner reads directions aloud

4. What to grade (lab points)

- Steps done (photo proof)
- Reflection: 2 sentences per box
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: Plug and air gap specs depend on the engines in Mr. Davis's shop; staff fill in the spec.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

⚡ Ignition Notes

Small Engines Overhaul · Unit 6 Ignition · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Magneto

1. Flywheel _____ pass the armature.
2. The armature makes high _____.
3. Small engines don't need a _____ for spark.

Word bank: **battery** **magnets** **voltage**

B. Spark plug

4. Match the part:

Terminal	spark starts
Insulator	spark jumps to
Center electrode	wire connects
Side electrode	white ceramic

C. Gaps and timing

5. The air gap is set with a gauge _____.
6. A sheared flywheel _____ throws off timing.

Word bank: **card** **key**

Level 2 version of the gen ed “Ignition Handout.”

📖 Ignition: Vocabulary Picture Cards (Level 2)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Spark plug

The part that makes a spark across its gap to light the air-fuel mix.



Magneto

The ignition system that uses flywheel magnets and a coil to make high voltage, no battery needed.



Armature (coil)

The coil mounted beside the flywheel that turns the magnets' energy into a high-voltage spark.



Air gap

The small space between the armature and the flywheel magnets, set with a gauge card.



Spark plug gap

The space between the plug's center and side electrodes, set with a gap gauge.



Flywheel key

A soft metal key that holds the flywheel in the right position for spark timing.



Kill switch

A switch or wire that grounds the coil to stop the spark and shut off the engine.



Spark tester

A tool clipped between the wire and ground to safely see if there is a spark.

⚡ Spark Plug & Spark Check

Small Engines Overhaul · Unit 6 · Level 2



Name: _____

Modified from the class “Ignition Handout”.

🧺 Supplies

- ☐ 🔧 **Spark plug socket** + ratchet
- ☐ 📏 **Gap gauge**
- ☐ 🧪 **Spark tester**
- ☐ 📄 **Gauge card** air gap
- ☐ ⚡ **New spark plug** optional
- ☐ 📋 **Plug color chart**

⚡ Steps: check each box when you finish

1. 🔌 Disconnect the plug wire. ☐
2. 🔧 Remove the spark plug with the socket. ☐
3. 📋 Compare the plug's color to the chart; record. ☐
4. 📏 Check and set the gap to spec. ☐
5. 🔩 Reinstall the plug by hand, then snug with the wrench. ☐
6. 🧪 Attach the spark tester; pull the rope; record spark color. ☐
7. 📄 With adult supervision, check the armature air gap with the gauge card. ☐

🚧 **Under construction:** Plug and air gap specs depend on the engines in Mr. Davis's shop; staff fill in the spec. *Check with the class teacher.*

⚡ Spark Plug & Spark Check · Prep Sheet

Small Engines Overhaul · Unit 6 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
🔧 **Mechanic**	Does the main engine job with the tools	
🧰 **Tool helper**	Gets tools from the board and returns them to staff	
📋 **Recorder**	Takes photos and fills in the chart	
🧽 **Clean-up captain**	Wipes the bench, sweeps, checks tools are back	

🧰 Tools and supplies: check when you have it

- ☐ 🔧 **Spark plug socket** + ratchet
- ☐ 📏 **Gap gauge**
- ☐ 🧪 **Spark tester**
- ☐ 📄 **Gauge card** air gap
- ☐ ⚡ **New spark plug** optional
- ☐ 📋 **Plug color chart**

⚠️ Safety and manners focus

🥽 **Safety glasses** on the whole time.

👖 **Coveralls** and closed-toe shoes; tie back long hair.

🔧 Use tools **with adult supervision**. Right tool for the job.

⚡ **Never hold the plug wire** while pulling: use the spark tester.

🧼 **Clean up** and wash hands when you finish.

✅ Before you start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

⚡ Spark Plug & Spark Check · Reflection

Small Engines Overhaul · Unit 6 · Level 2



Name: _____

📸 My photos / proof

☐ Before photo ☐ During photo ☐ Finished photo

✅ Skills I used today (check all you did)

☐ 🥽 Wore safety glasses and used tools safely

☐ 🔧 Finished the Spark Plug & Spark Check steps

☐ 🧽 Cleaned up and returned tools

☐ 🤝 Did my job role

What did you do today, in order?

1. _____
2. _____

Ideas: **First…** **Next…** **Last…**

What did you learn about engines?

1. _____
2. _____

Ideas: **The engine needs…** **I saw…** **Next time…**

✅ Ignition Quiz

Small Engines Overhaul · Unit 6 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🧲 A magneto makes electricity with…

A. A battery | B. Magnets and a coil | C. Gasoline

2. 🔘 The magnets are in the…

A. Carburetor | B. Muffler | C. Flywheel

3. 🌀 The armature is…

A. The coil beside the flywheel | B. The fuel bowl | C. A valve

4. ⚡ The spark jumps from the center electrode to the…

A. Gas tank | B. Piston ring | C. Side electrode

5. 📏 Spark plug gap is set with a…

A. Gap gauge | B. Hammer | C. Ruler

6. ↔️ Armature air gap is between the coil and the…

A. Muffler | B. Flywheel | C. Fuel tank

7. 🔘 A sheared flywheel key causes…

A. More power | B. Bad timing, hard starting | C. Less noise

8. 🔌 The kill switch stops the engine by…

A. Grounding the coil | B. Draining fuel | C. Removing oil

9. 🧪 A good spark is…

A. Weak orange | B. None | C. Bright blue

10. 🔌 Before testing, you…

A. Disconnect the plug wire | B. Add gas | C. Run it indoors

⚡ Ignition Notes · ANSWER KEY

Small Engines Overhaul · Unit 6 Ignition · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Magneto

1. Flywheel **magnets** pass the armature.
2. The armature makes high **voltage** .
3. Small engines don't need a **battery** for spark.

Word bank: **battery** **magnets** **voltage**

B. Spark plug

4. Match the part:

(Key: Terminal → wire connects · Insulator → white ceramic · Center electrode → spark starts · Side electrode → spark jumps to)

Terminal	spark starts
Insulator	spark jumps to
Center electrode	wire connects
Side electrode	white ceramic

C. Gaps and timing

5. The air gap is set with a gauge **card** .
6. A sheared flywheel **key** throws off timing.

Word bank: **card** **key**

Level 2 version of the gen ed "Ignition Handout."
🔑 Key written by our team from the class teacher's materials, please check.

✅ Ignition Quiz · ANSWER KEY

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Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🧲 A magneto makes electricity with… Ignition

A. A battery | **B. Magnets and a coil** | C. Gasoline

2. 🔘 The magnets are in the… Ignition

A. Carburetor | B. Muffler | **C. Flywheel**

3. 🌀 The armature is… Ignition

A. The coil beside the flywheel | B. The fuel bowl | C. A valve

4. ⚡ The spark jumps from the center electrode to the… Ignition

A. Gas tank | B. Piston ring | **C. Side electrode**

5. 📏 Spark plug gap is set with a… Ignition

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7. 🔘 A sheared flywheel key causes… Ignition

A. More power | **B. Bad timing, hard starting** | C. Less noise

8. 🔌 The kill switch stops the engine by… Ignition

A. Grounding the coil | B. Draining fuel | C. Removing oil

9. 🧪 A good spark is… Ignition

A. Weak orange | B. None | **C. Bright blue**

10. 🔌 Before testing, you… Ignition

A. Disconnect the plug wire | B. Add gas | C. Run it indoors

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed “Ignition Handout (title only; team-written)”. Score: __ / 10. 🔑 Key written by our team from Mr. Davis's materials, please check.